



Adaptive Neural Control for Mars Rover Wheel Entrapment Recovery in Granular Terrain

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The Problem That Started It All

The Spirit Rover Incident: May 2009

- After **5 years** of exploration, Spirit drove into soft sulfate-rich regolith at “Troy”
- **8 months** of Earth-based recovery attempts, all failed
- One-way signal latency: **4–24 minutes**
- No autonomous escape capability onboard
- Mission permanently ended; rover never moved again

*“What if the rover could have **saved itself**?”*

Why Autonomous Recovery?

-  Real-time teleoperation: **impossible** at 4–24 min latency
-  Rocker-bogie suspension: complex sinkage dynamics
-  Martian regolith: highly unpredictable granular behaviour
-  Extended missions encounter unanticipated terrain types

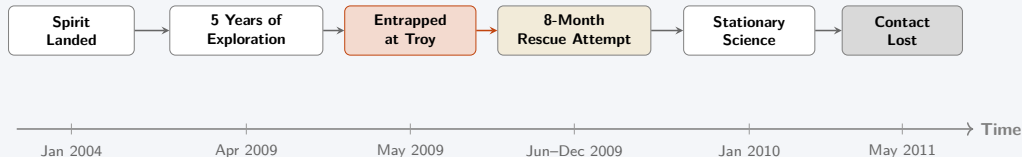


150M km

Earth–Mars distance during entrapment

The Spirit Rover Entrapment: A 7-Year Timeline

🕒 Chronology of the Spirit Entrapment Incident (2004–2011)



Spirit's wheels broke through a thin crusty surface into soft sulfate-rich soil. Despite 8 months of ground-based testing and careful extraction commands, the rover remained permanently trapped, ending its mobile mission.

Key lesson: Spirit had *no onboard capability* to detect or respond to entrapment autonomously. All recovery attempts were commanded from Earth with 8–40 min round-trip delay. **Our work builds the autonomy Spirit never had.**

Research Objectives

Obj 1: Realistic Simulation

Couple **Newton MPM** granular physics with **Isaac Lab** rigid-body solver under **Martian gravity** (3.72 m/s^2)

Obj 2: Entrapment Detection

Dual-sensor scheme: **slip-ratio monitoring** + **drive-torque anomaly**: no GPS, no external sensing

Obj 3: Recovery Policy

Recurrent RL policy (**PPO + GRU**) discovering rocking, steering pivots, and differential drive for varying sinkage

Obj 4: Curriculum Learning

Progressive sinkage: **12 cm** → **22 cm** across 64 parallel environments; enables generalisation across terrain conditions

Obj 5: Sim-to-Sim & Sim-to-Real

Primary: Cross-simulator validation (MuJoCo / Gazebo), sloped beds $\pm 15^\circ$, variable soil density

Secondary: ONNX export → **Raspberry Pi 5** at 10 Hz

✓ **Novel contribution:** First DRL escape system for a 6-wheel rocker-bogie rover in physics-accurate granular simulation

Related Work & Research Gaps

Work	Robot	Key Limitation	Terrain	Gravity	Escape%
Bi & Ding (2026)	4-wheel SSMR	No granular physics; needs expert demos	Rigid	Earth	90%
Mortensen & Bøgh (2024)	AAU 6-wheel	Navigation only; no entrapment/recovery task	Rigid	Mars	N/A
Inotsume (2020)	6-wheel	Detection only; no recovery policy	Bekker	Mars	N/A
Jain (2019)	6-wheel	Discrete action space; approx. terrain	Approx.	Mars	N/A
Fankhauser (2018)	ANYmal	Legged robot; different failure mode	Rigid	Earth	N/A
Ours	6-wheel RB	Early training; sim-to-real pending	MPM	Mars	~35%



Key Dependency: RL RoverLab (Mortensen & Bøgh, ISPaRo 2024)

Open-source RL suite on NVIDIA ORBIT/Isaac Sim. Provides the **AAU Rover Complex** USD asset and pre-trained **navigation policies** (our Navigation Brain). **Gap we fill:** rigid terrain only; no granular physics, no entrapment task. We add Newton MPM coupling for regolith sinkage and recovery.



Gap 1

No prior work: 6-wheel rocker-bogie + MPM granular + DRL



Gap 2

Granular progressive sinkage unstudied in DRL context



Gap 3

Mars gravity + independent 10D per-wheel control unexplored

Comparison with State-of-the-Art: Bi & Ding (2026)

Aspect	Bi & Ding (2026)	Ours
Robot	4-wheel SSMR	6-wheel RB
Terrain	Rigid (Unity)	MPM (Newton)
Gravity	Earth	Mars
Obs	30D	29D
Action	2D	10D
Policy Steps	PPO+LSTM 4M	PPO+GRU 200K
Reward	GAIL+expert	Handcrafted
Escape	90%	~35%
Real-world	Yes	Planned

⚠️ **Their edge:** 20× more training, real-world validated, GAIL+expert demos

✅ **Our edge:** physics-accurate MPM soil, Mars gravity, 5× richer action space, no expert data

📖 Why GRU not LSTM?

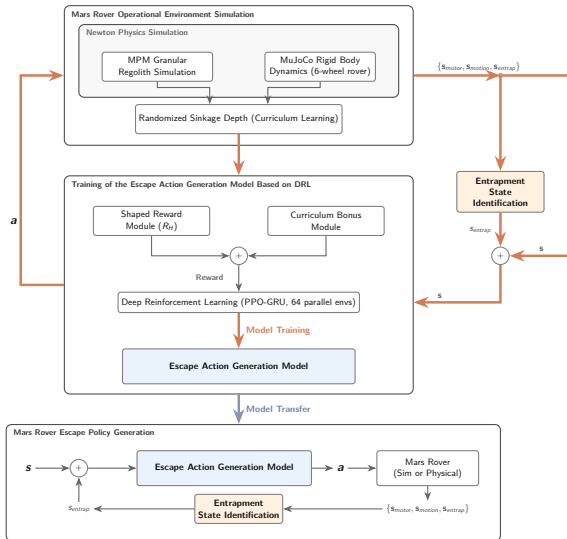
GRU advantages:

- 2 gates vs. 3 in LSTM; ~30% fewer params
- Faster on RPi5 (<5 ms vs. ~7 ms)
- No cell state; simpler BPTT

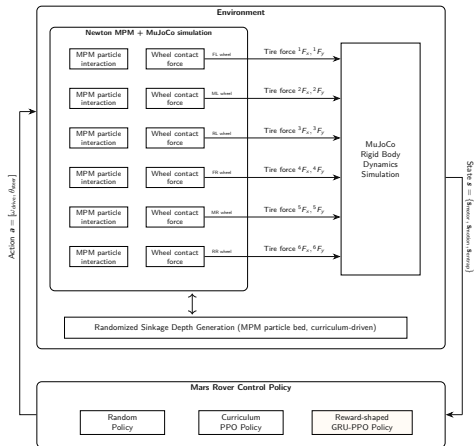
LSTM not needed:

- Episodes 200–600 steps, seq=16; no very long dependencies
- GRU matches LSTM for sequences ≤5 s
- PPO clipping already stabilises gradients

Escape Policy Generation: Training & Deployment (Fig. 2)

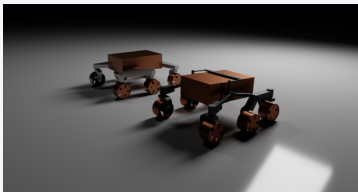


Simulation Framework: Fig. 3



Robot Platform: AAU 6-Wheel Rover from RL RoverLab

AAU Mars Rover USD Asset (RLRoverLab)



About RLRoverLab

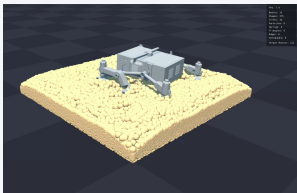
Open-source Isaac Lab rover framework by Magni et al.

Provides **USD assets** for multiple planetary rovers including the AAU 6-wheeled rocker-bogie

Includes **navigation policies** for point-to-point traversal; our **Navigation Brain** in the dual-brain architecture

We **extend** RLRoverLab by coupling Newton MPM granular terrain to enable entrapment simulation

Rover on MPM Regolith Bed (Newton Sim)



AAU Rover Joint Configuration

Joint type	Count	Range
Drive (continuous)	6	± 6 rad/s
Steer (revolute)	4	± 0.6 rad
Passive (rocker/diff.)	N	Free

10D action space: 6 independent drive ω + 4 steer θ , the richest per-wheel control in any entrapment DRL work

Physics Simulation Stack: Newton MPM

Material Point Method (MPM)

Hybrid **Eulerian–Lagrangian** continuum method
Particles track mass, momentum, deformation gradient

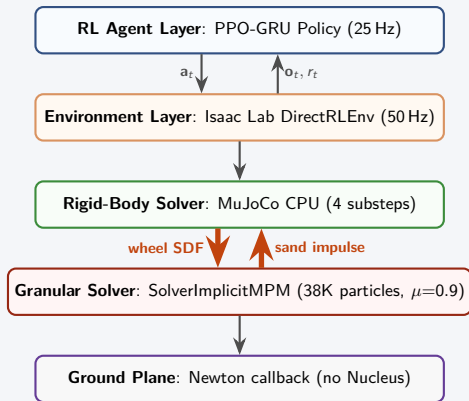
Drucker-Prager yield surface for granular flow
Correctly models: sinkage progression, bulldozing resistance, particle redistribution during wheel spin
38,400 particles per environment

MPM vs. rigid approx: **physics-accurate**
traction & sinkage

Sand Bed Configuration

Bed size	2.0 m × 2.0 m × 0.25 m
Voxel size	0.05 m grid resolution
Friction μ	0.9 (wheel–sand)
Gravity	(0, 0, -3.72) m/s ²
Solver	SolverImplicitMPM
Substeps	4 per physics step
Transfer	APIC (affine particle-in-cell)

Physics Simulation Stack (Fig. 7)



64 parallel envs · **38K** MPM particles each

RL Formulation: POMDP & Reward Design

POMDP Formulation

Entrapment recovery cast as a **POMDP**:

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right], \quad \gamma = 0.99$$

Partial observability: rover cannot directly sense per-wheel sinkage, terrain composition, entrapment boundary

→ GRU hidden state h_t **implicitly estimates** unobserved variables from observation history

PPO Clipped Objective

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]$$

Clip ratio $\epsilon = 0.2$, GAE $\lambda = 0.95$

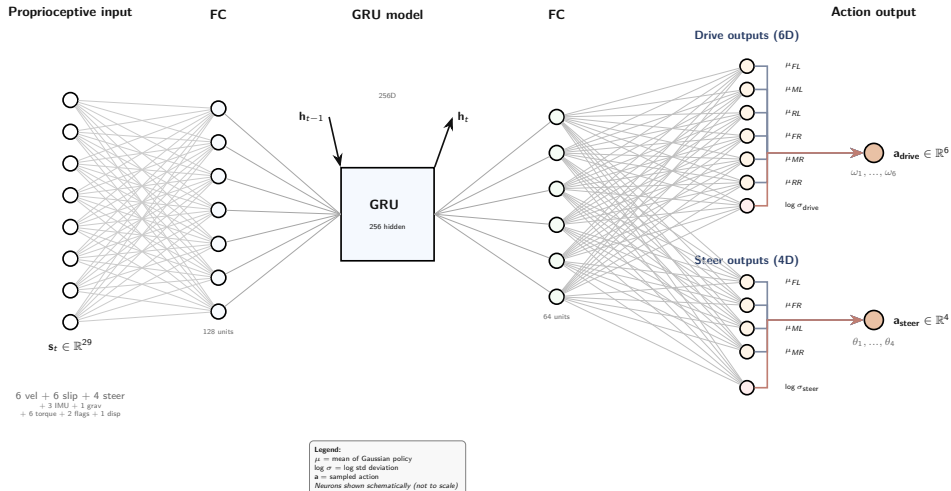
$$R_t = r_{\text{progress}} + r_{\text{escape}} + r_{\text{rocking}} - p_{\text{slip}} - p_{\text{tilt}} - p_{\text{smooth}} - p_{\text{abnormal}}$$

Reward Function R_t (v4)

Term	Weight	Purpose
r_{progress}	+1.5	Forward velocity (suppressed when trapped)
r_{escape}	+20	Sparse milestones at 0.5/1.0/1.5 m
r_{rocking}	+2.0	$\Delta v_x \cdot f_{\text{ent}}$, active rocking
p_{slip}	-0.5	Excessive wheel spin
p_{tilt}	-0.3	Roll/pitch instability
p_{smooth}	-0.05	Jerky action changes
p_{abnormal}	-0.3	Backward motion under anomaly

Key: r_{escape} (w=20) dominates during entrapment;
 r_{rocking} incentivises active self-extraction

GRU-PPO Policy Network Architecture



Dual-Sensor Entrapment Detection

Formal Entrapment Criterion

Entrapment flag $f_{\text{ent}} = 1$ iff **all conditions** hold for ≥ 15 consecutive steps (0.6 s):

$$f_{\text{ent}} = 1 \iff \begin{cases} |v_x| < 0.15 \text{ m/s} \\ \bar{\sigma}_{\text{slip}} > 0.4 \end{cases}$$

where $\bar{\sigma} = \frac{1}{6} \sum_i \sigma_i$

Note: f_{ent} is pre-set to 1 at episode reset (rover spawns already buried 12–22 cm)

Dual Detection Mechanisms

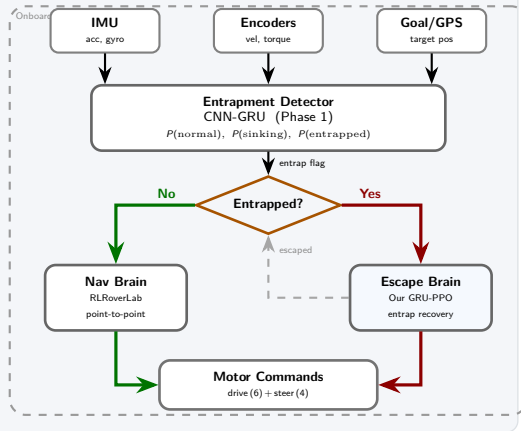
Slip-based:

$$f_{\text{ent}} = \mathbb{I}[|v_x| < 0.15 \wedge \bar{\sigma} > 0.4 \text{ for 15 steps}]$$

Torque-based:

$$f_{\text{anom}} = \mathbb{I}[\overline{|\tau_i / \tau_{\text{max}}|} > 0.85 \wedge \Delta \bar{\tau} > 0.10 \text{ for 20 steps}]$$

Dual-Brain Onboard Architecture (Fig. 11)



Curriculum Learning & Training Pipeline

Progressive Sinking Curriculum

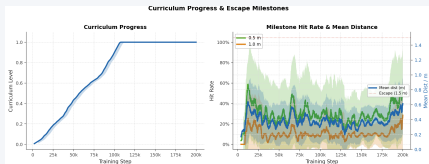
$$d_{\text{sinking}}(p) = d_{\min} + p \cdot (d_{\max} - d_{\min})$$

$$d_{\min} = 0.12 \text{ m}, \quad d_{\max} = 0.22 \text{ m}$$
$$p \in [0, 1] \text{ (curriculum progress)}$$

🔧 PPO Hyperparameters

Rollout length	24 steps	Learning rate	3×10^{-4}
Epochs / rollout	5	Clip ϵ	0.2
Mini-batches	4	Discount γ	0.99
Entropy coeff	0.03	GAE λ	0.95
GRU hidden	256	BPTT seq	16

📈 Curriculum Progress (Real Training Data)

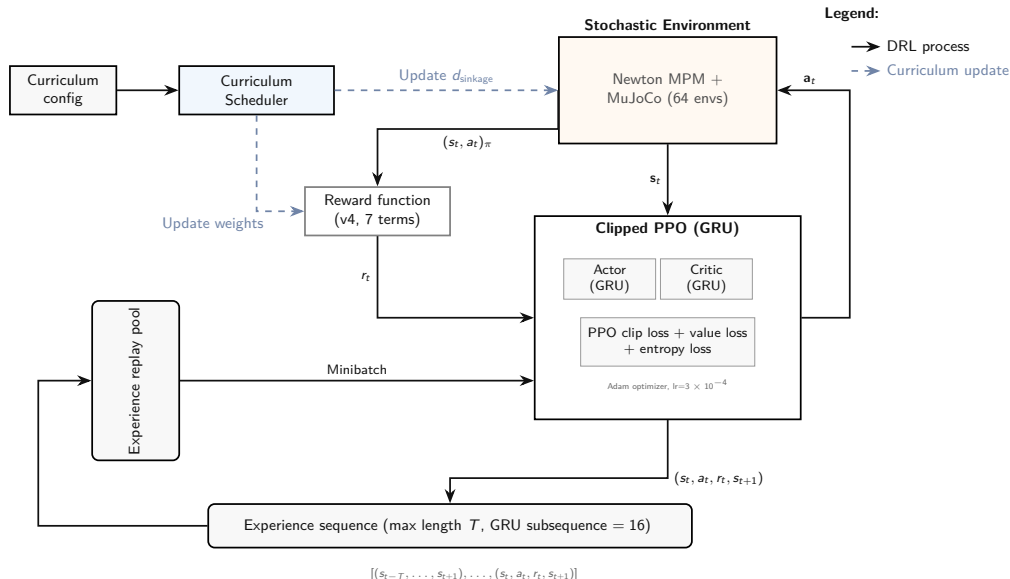


☰ Training Hardware

GPU	NVIDIA RTX 5070 Ti (16 GB GDDR7)
CPU	Intel i7-1255U, 10 cores @ 4.7 GHz
RAM	40 GB DDR5
Envs	64 parallel (32 during debug)
Wall time	~6–8 hrs for 200K timesteps

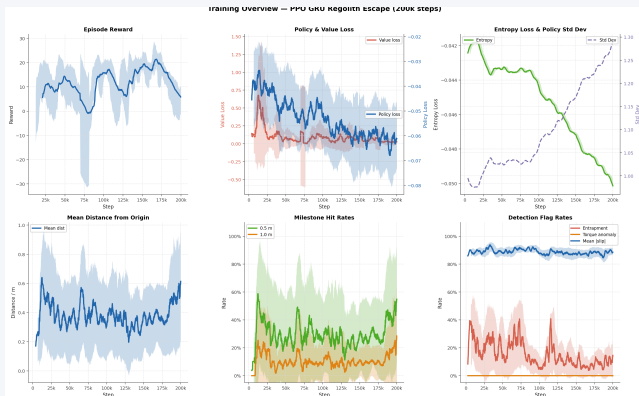
Domain randomization: motor gain $\mathcal{U}(0.8, 1.2)$, obs. noise $\mathcal{N}(0, 0.02)$, spawn yaw $\pm 30^\circ$

DRL Training Loop: PPO-GRU with Curriculum Scheduler



Training Overview: 200K Timesteps

Training Dashboard (PPO-GRU, 200K steps)



~35%

0.5 m milestone hit rate

(15% at 1.0 m)

7.8

Mean episode reward

(last 10% of run)

1.0

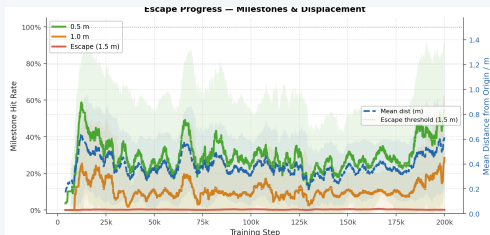
Curriculum progress

(reached at ~110K) ✓

Policy std dev: 1.303, still exploring; 2M+ steps planned

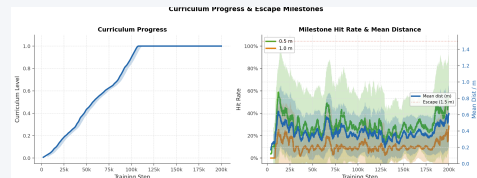
Escape Progress & Curriculum Dynamics

🎯 Milestone Hit Rate & Mean Displacement



0.5 m milestone hits ~35%
smoothed mean; rover escapes first
hurdle reliably

📈 Curriculum Progress & Sinkage Milestones



Curriculum saturates at step ~110K
, full sinkage range (22 cm)
maintained thereafter

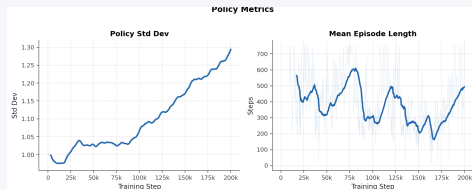
1.5 m escape never triggered
, episode resets after 0.5 m clear
(correct behaviour, not failure)

Loss & Policy Metrics

Training Losses



Policy Std Dev & Episode Length



GRU-PPO vs. Feedforward: Architecture Comparison

Feature	GRU-PPO	FF-PPO
Memory depth	256 hidden	None
Sinkage tracking	✓	✗
Parameters	~200K	~50K
RPi5 inference	<5 ms	<2 ms

⚠ Key Training Observations

Value loss spikes at 25K (new terrain difficulty), then converges to ~ 0

Policy loss drifts $-0.02 \rightarrow -0.08$, healthy gradient signal

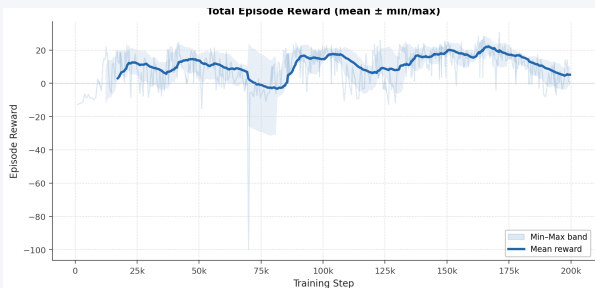
Entropy loss declining: policy concentrating without collapse

Std dev rising $1.0 \rightarrow 1.3$: still in exploratory regime at 200K

Episode length 200–600 steps: varied escape speeds emerging

Reward Component Breakdown

Per-Component Reward Contributions



! Why $p_{\text{abnormal}} \approx 0$?

Fires **only if both hold ≥ 20 steps**: (1) avg. torque $> 85\%$ of limit ($\overline{|\tau / \tau_{\text{lim}}|} > 0.85$) **AND** (2) torque rapidly changing ($\Delta \bar{\tau} > 0.10$). Policy avoids saturating motors, so condition 1 is almost never met.

Fixes: use OR logic; lower threshold to 0.70; shorten window to 10 steps; or replace with soft penalty $p = -0.3 \cdot |\tau / \tau_{\text{lim}}|$.

i Reward Analysis

r_{escape} ($w=20$) is dominant positive signal, milestone-driven

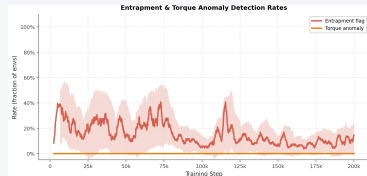
r_{rocking} ($w=2$) activates only when $f_{\text{ent}} = 1$

p_{slip} (0.5) suppressed during entrapment, active post-escape

p_{abnormal} (-0.3) ≈ 0 : dual-condition gate rarely satisfied

Net reward positive throughout: policy avoids catastrophic actions

🚩 Entrapment & Anomaly Detection Flags



Escape Behaviour: Live Simulation Demo



Click to play (Adobe Acrobat Reader)

Other viewers: youtu.be/LUWEIS6d7CQ

Emergent Escape Strategies

1. **Rocking**: alternating fwd/bwd to compact sand beneath wheels
2. **Differential steering**: yaw toward traction-rich zones
3. **Coordinated steer-and-drive**: diagonal escape paths
4. **Progressive acceleration**: avoid deeper re-sinkage

All strategies emerged without explicit programming

Environment Metrics



Training Results: Summary

Metric	Achieved	Target	Status
Reward (last 10%)	7.8	> 0	✓ Positive
0.5 m escape rate	~35%	>50%	↑ Progress
1.0 m escape rate	~15%	>30%	↑ Progress
Curriculum progress	1.0	1.0	✓ Complete
Mean v_x	0.12 m/s	> 0.10	✓ Achieved
Policy std dev	1.303	<1.0	— Converging

⚠ **Requires 2M+ timesteps** for stable >50% escape.
Planned: RTX 4090, 512 envs, 10× scale-up

★ What Worked Well

- ✓ Curriculum saturated to full 22 cm sinkage
- ✓ Positive episode reward throughout
- ✓ Emergent rocking & steering strategies
- ✓ Policy entropy stable (no collapse)
- ✓ Value loss converged cleanly post-25K

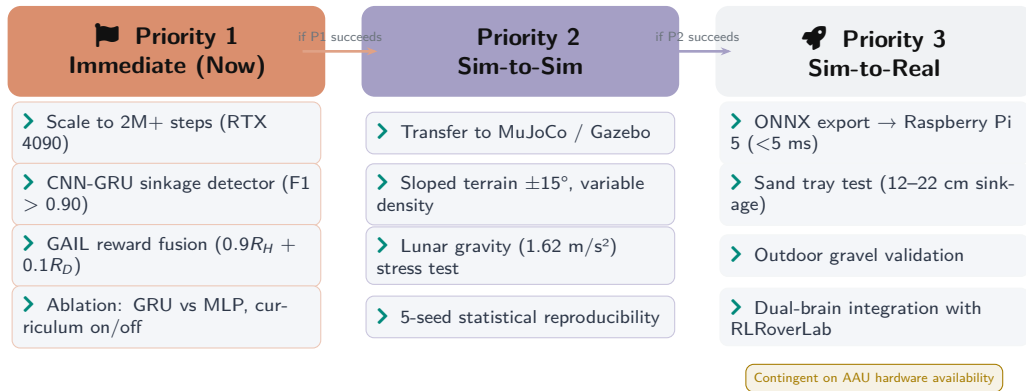
⚠ Known Limitations

- ⚠ 200K steps insufficient for full convergence
- ⚠ Torque anomaly detector fires <1% (dual-condition too strict)
- ⚠ Episode resets before 1.5 m milestone
- ⚠ No feedforward baseline comparison yet

📊 Summary Dashboard

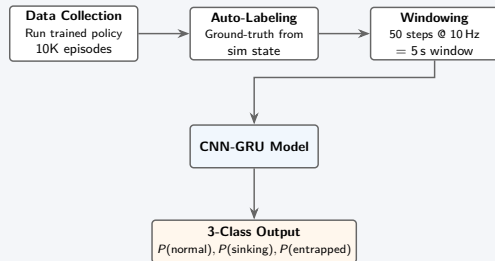
Training Summary				
Time	Steps	Reward (last 10%)	Min. Reward	Policy Std Dev
ppo_gnn_reg30th	195,600	7.80	99.97	1.303

Future Work Roadmap



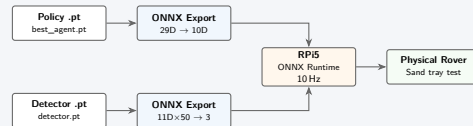
Future Pipelines: Detection & Sim-to-Real

Phase 2: CNN-GRU Sinkage Detection Pipeline



Target: $F1 > 0.90$ on “entrapped” class, latency < 1 s,
inference < 5 ms on RPi5

Phase 3: Sim-to-Real Deployment Pipeline



Target: inference < 5 ms per step on RPi5 via ONNX

Runtime (no GPU)

Summary of Contributions

★ Technical Contributions

1. **First** Newton MPM + Isaac Lab integration for rover entrapment simulation under Martian gravity
2. **Novel** dual-sensor detection: slip-ratio + torque anomaly flagging without external sensing
3. **Recurrent** GRU-PPO policy enabling multi-step memory-based escape strategies
4. **Seven-component** shaped reward with progressive sinkage curriculum (12–22 cm)
5. **Complete pipeline**: physics sim → RL training → evaluation → visualization dashboard

🔍 Research Gaps Addressed

- ✓ 6-wheel rocker-bogie + MPM granular + DRL (no prior work)
- ✓ Granular progressive sinkage as DRL entrapment mode (first study)
- ✓ Mars gravity effect on traction in DRL recovery (no prior work)
- ✓ 10D per-wheel action space (vs 2D in all prior work)
- ✓ Curriculum-driven sinkage depth progression

Physics-accurate · Mars-gravity-aware ·
Curriculum-trained · 6-wheel DRL Recovery



Thank You!



Questions & Discussion

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📺 [Demo:youtu.be/LUWEIS6d7CQ](https://youtu.be/LUWEIS6d7CQ)